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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Driven Self-Adaptive Caching for Dynamic Web Application Performance	
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Abstract

AI-Driven Self-Adaptive Caching for Dynamic Web Application Performance is a project that aims to improve the performance, scalability, and responsiveness of modern dynamic web applications through intelligent and adaptive caching techniques. Dynamic web applications frequently handle changing user requests, varying traffic patterns, and continuously updated content. Traditional caching mechanisms generally depend on static rules, fixed expiration times, or manually configured policies, which may not efficiently respond to these changes. As a result, applications may experience increased response time, higher server load, unnecessary database requests, and reduced overall performance.

The proposed system introduces an AI-driven self-adaptive caching mechanism that continuously monitors application behavior and analyzes factors such as request frequency, content popularity, access patterns, cache hit and miss rates, and system workload. Based on these observations, the system intelligently determines which resources should be cached, retained, updated, or removed. Machine learning techniques can be used to predict frequently requested resources and identify changing access patterns, allowing the caching strategy to automatically adapt without requiring constant manual configuration.

The system is designed to dynamically modify caching policies according to real-time workload conditions. Frequently accessed and predictable resources can be prioritized for caching, while less useful or outdated resources can be removed to optimize available cache space. This adaptive approach can reduce database access, network overhead, and processing requirements while improving application responsiveness.

The effectiveness of the proposed system can be evaluated using performance metrics such as cache hit ratio, response time, latency, throughput, database request reduction, and server resource utilization. The results can be compared with conventional caching approaches to determine the benefits of AI-based

adaptation. The project demonstrates how Artificial Intelligence and self-adaptive computing can be integrated into web application architectures to create efficient, scalable, and intelligent caching systems capable of responding to changing application workloads.

Signature of the Guide HOD